Element A and B google doc:

Students: 001, 012, 013

A1- A main problem we have discovered with the mini golf course is strange elevation changes that are in the area we are building on. In the area that we are building the mini golf course there are many obstacles. Some include a tree, drainage systems, and varying elevations. Another problem would be the budget we have to work with. 10,000 dollars seems like alot, however things like building materials are expensive. A problem we also might have is the basic designing, our design needs to hold up against storms and weather, but also against the many students that will be using the course. It also needs to be safe for students but also passing cars.

A2- Our first problem, which is elevation changes, can be defined as problems with the land. The land outside has very few flat places and will be extremely difficult to build on. The slopes and elevation changes would take a lot of money and time to fix or even just adjust, it also would be alot of construction outside and it could affect morning and afternoon traffic since it's that close to the road. Our second problem is our budget. Our course would need a lot of things that add up to a lot of money. Things like turf, something to keep people and golf balls away from the roads, and the building materials for the course itself, and a construction team. All of these are large expenses and we don't have much money to use. And our last problem listed is the safety concerns. Teenagers and kids are irresponsible, with having a mini golf course specifically made for kids that close to a road is dangerous. It's dangerous not only for the kids playing but also the cars passing by. Also if the design is breaking overtime due to not holding up against storms it also becomes a safety hazard.

A3- If we figure out how to solve the elevation problems or use it to our advantage we can in return help other people build things on areas they didn't think were possible to build on due to strange elevation.This leads to more opportunities for people to use the land they have in a cost efficient way. The budget could also be used as a way to show other school districts that even though there may be a strict budget, there are many ways to use money efficiently. And finally, if we can figure out design safety along with our budget then we can show other schools or organizations that building a safe and fun thing like a mini golf course is possible. Many other organizations can take our designs and use them for their own use, places like nursing homes and hospitals.

A4- Some of the primary stakeholders include elementary administrators and students, highschool administrators and students.The secondary stakeholders include the community, which is people that live or travel near the school. The concerns from the primary stakeholders, starting with administrators, may include: Is the design safe? How will this affect the education of our students? Will it keep students entertained for a while? Is this design age appropriate for the students?. Questions like these from our administrators cause the need for a safe, age appropriate, and engaging design. Our secondary stakeholders, the students, may have questions like: Will the course be used during most school days or only for special events? Is the course going to keep me from learning? Will highschool students and elementary students use the course at the same time?. Questions like these from our secondary stakeholders cause the need for more answers about the times the course will be used. It could also cause the need for answers on how the golf course can positively or negatively affect the education of students. These questions need answers before there is any way of creating a final design. We need to know the needs and wants of the students and faculty so we can construct the best course geared towards the people.

A5- The first problem, elevation changes, can be justified by the fact that mini golf is better played on a flat surface. However, to keep the players engaged, elevation changes add difficulty, and the longer it takes to complete the course, the longer the students are engaging in physical activities. The next problem, the budget, is justified by the cost of building materials alone. The cost for turf for a mini golf course could easily be 5-20 dollars per square foot, and the location where we are putting this is pretty large. Our last problem, basic designing, can be justified by the fact that safety is most stakeholders' biggest concern. The basic design of the minigolf course has to be not only budget friendly but also safe to the students and cars passing by. Teenagers have a tendency to be immature and act irresponsibly, the minigolf course being near a main road as well as

between two parking lots is dangerous. If an accident were to happen to a student or vehicle while playing, the course could end up costing the school a lot of money. That would lead to administrators not letting students choose what to do with the money next time.
A6- The elevation change is important because if it is not solved correctly the golf course design may fail. Uneven grounds can cause the ball to roll where it's extremely difficult to get it in the hole, or even where it's not supposed to be. The budget is important because if we started construction on a golf course we can't afford then we would end up with an unfinished course. Also if we dont budget responsibly then we could end up never being allowed to use grant money on our own designs again. And finally safety is the most important, if a student or staff member is injured on the golf course, it could get the school in alot of trouble. The safety of students is something that should be our biggest priority. And if the students end up damaging the cars driving by or parked, then that can also cost the school a lot of money and possibly the student.

Element B:

B1- After long research behind any prior attempts at building an outdoor minigolf course, my group could not find anything. We found very similar attempts like at Shawnee Highschool where they built an outdoor course. But nothing that really matched the criteria we had and the problems we need to work around.